

Architecting Secure Multi-Agent Intelligence for Illiquid Financial Ecosystems: A Temporal Fusion Approach for North African Equity Markets

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Abstract—Financial AI systems are predominantly designed and evaluated on liquid, developed markets. Thin, illiquid emerging markets present distinct challenges: heterogeneous feature scales across stocks, sparse volume signals, limited temporal coverage, and no public benchmarks. This paper presents a seven-agent architecture combining Temporal Fusion Transformers, XGBoost fundamental analysis, multilingual NLP sentiment scoring, Graph Attention Networks, and explainable AI components. The system is designed, trained, and evaluated on a real emerging market dataset spanning 10 years (2016–2025) with 68 stocks and 116,626 trading records. The Temporal Fusion Transformer achieves 77.4% directional accuracy on unseen 2025 test data, representing a 25.5% improvement over the random baseline. An empirical ablation study across four CNN-LSTM configurations establishes an architectural ceiling of 53.8%, revealing that convolutional approaches are fundamentally unsuited to heterogeneous financial feature matrices. Per-stock normalization and confidence-gated agent isolation provide robustness against data poisoning in thin markets. The architecture satisfies EU AI Act transparency requirements through integrated explainability and produces an auditable decision trace for regulatory compliance. This work demonstrates that attention-based architectures with variable selection networks outperform conventional convolutional-recurrent models on emerging market data, and that multi-agent systems with adaptive weighting can maintain prediction quality under heterogeneous market conditions.

Index Terms—Temporal Fusion Transformers, multi-agent systems, emerging markets, explainable AI, financial forecasting, graph neural networks, thin liquidity, EU AI Act compliance

I. INTRODUCTION

A. Global Context: AI in Financial Markets

Since 2020, artificial intelligence has transformed global financial markets. As of 2023, over 70% of trading volume on major exchanges (NYSE, NASDAQ, LSE) is driven by algorithmic systems. However, these systems are overwhelmingly designed and evaluated on liquid developed markets with continuous trading, deep order books, high daily volume, and abundant historical data.

When deployed to emerging markets with thin liquidity, these models fail dramatically. Thin markets are characterized by:

- **Price continuity gaps:** stocks that do not trade for days or weeks
- **Heterogeneous feature scales:** stock prices range from 1 Dinar to 50 Dinar; daily volume from 50 to 50,000 shares — all within the same market

- **Absence of derivatives:** no options, futures, or synthetic instruments that developed market models rely on
- **Limited linguistic diversity:** news predominantly in French and Arabic, creating NLP challenges
- **No public benchmarks:** no standard datasets or baseline models exist for these markets

Global AI systems trained on S&P 500 or NASDAQ data cannot generalize to these conditions. The assumptions embedded in their architectures — global normalization across all stocks, dense feature coverage, high-volume price signals — are violated systematically in thin markets.

B. Challenges of Emerging Market Prediction

The Tunisian stock exchange (BVMT) serves as a representative testbed for these challenges. With 68 listed securities and average daily trading of 500,000–2,000,000 Dinars, the BVMT experiences:

- Trading halts due to low volume (price suspension days)
- Extreme volatility spikes when volume appears after days of no trading
- Company-identity signals embedded in absolute price levels (AMEN BANK perpetually trades at ~22 DT; BH LEASING at ~1.5 DT)
- Limited financial data: quarterly and annual reports rather than real-time earnings guidance
- News predominantly in French and Arabic with domain-specific terminology not well-represented in general-purpose language models

This market presents both challenges and opportunities. The absence of existing AI systems means the research contribution is not merely an incremental improvement to a known system, but rather the first rigorous scientific evaluation of multi-agent AI architectures on such a market.

C. Research Contributions

This paper makes the following contributions:

- 1) **Temporal Fusion Transformer adaptation for thin markets:** A TFT model architecture incorporating per-stock normalization, confidence-gated feature selection, and multi-horizon quantile regression, achieving 77.4% directional accuracy on seven-day forecasts. The design explicitly handles heterogeneous data scales and sparse volume signals through the Variable Selection Network's learned feature importance weights.

- 2) **Empirical architectural evaluation:** A rigorous five-run comparison study establishing that CNN-LSTM architectures achieve a ceiling of 53.8% accuracy regardless of hyperparameter tuning. Root cause analysis reveals that convolutional kernels treat heterogeneous financial features (price, RSI, volume) as spatially equivalent — a fundamental architectural mismatch for this data structure. This is the first explicit architectural comparison confirming TFT’s superiority on emerging market data.
- 3) **Multi-agent ensemble with failure isolation and confidence gating:** A seven-agent architecture implementing agent-level trust boundaries, confidence-gated weight adaptation, and failure isolation. The design provides robustness against data poisoning in thin markets where individual manipulated trades can shift global statistics.
- 4) **Explainability framework satisfying EU AI Act requirements:** An integrated XAI layer producing SHAP-based feature importance, attention weight extraction, and plain-language decision explanations that satisfy Articles 13 and 14 of the EU AI Act (2024) transparency requirements for high-risk AI in finance.
- 5) **Graph-based market surveillance framework:** A framework for detecting anomalous stock decoupling using rolling correlation networks, with application to market integrity monitoring and insider trading detection.

D. Paper Organization

The remainder of this paper is structured as follows. Section II reviews related work on temporal models for financial forecasting, CNN-LSTM hybrid architectures, multi-agent systems in finance, explainable AI for financial compliance, graph neural networks for market structure analysis, and NLP for non-English financial markets. Section III presents the seven-agent system architecture and describes agent communication protocols, failure isolation, adaptive weight matrices, and confidence gating mechanisms. Section IV details the dataset construction, feature engineering process, TFT model configuration including the Variable Selection Network formulation, walk-forward validation protocol, and CNN-LSTM comparison architecture. Section V presents experimental results including CNN-LSTM ablation studies, TFT version progression, and literature context. Section VI discusses the 77.4% directional accuracy result, interprets the 25.5% improvement over baseline in the context of market class imbalance, presents information-theoretic analysis of loss improvements, and discusses quantile regression uncertainty quantification. Section VII presents security and privacy implications including adversarial robustness in thin markets, XAI as audit trails, agent trust and isolation properties, and graph-based market surveillance applications. Section VIII concludes with limitations and future work.

II. RELATED WORK

A. Temporal Models for Financial Forecasting

Time series forecasting in finance traditionally relied on ARIMA and exponential smoothing approaches. Modern deep learning has shifted the landscape toward recurrent neural networks and attention mechanisms.

LSTMs and their variants became the standard for financial time series due to their ability to capture long-range dependencies. However, LSTMs suffer from limitations on multi-variate, heterogeneous data. Specifically, LSTMs process a single hidden state through all input dimensions, forcing the network to compress information about price, volume, and momentum into a shared representation. This leads to representation bottlenecks when features have fundamentally different scales and semantics.

Attention mechanisms address this through explicit feature weighting. The Transformer architecture (Vaswani et al., 2017) introduced multi-head self-attention, allowing the model to attend to different representation subspaces. The Temporal Fusion Transformer (Lim et al., 2019) extends this specifically for time series: it adds a Variable Selection Network that learns per-feature importance weights **before** temporal processing. This is critical for heterogeneous financial data where feature relevance varies dramatically across assets.

Most published TFT applications target developed markets (S&P 500, NASDAQ, European indices) with reported directional accuracy ranges of 55–65% on daily forecasts. This paper demonstrates that with careful adaptation to thin market conditions — specifically per-stock normalization and confidence gating — TFT can achieve 77.4% on emerging market data with an equivalent or longer prediction horizon (seven days).

B. CNN-LSTM Hybrid Architectures

CNN-LSTM hybrids have been explored for financial forecasting as a way to combine spatial and temporal pattern detection. The motivation is sound: CNNs efficiently extract local patterns through convolutional filters, and LSTMs process temporal sequences.

However, a critical assumption underlying CNN-LSTM architectures is that input dimensions are spatially correlated — that adjacent columns (or nearby positions in a 2D matrix) encode related information. This assumption is valid for image data where adjacent pixels are correlated. For financial feature matrices of shape [60 days, 15 features], this assumption is violated: `close_price`, `RSI_14`, and `volume_MA_20` are fundamentally different quantities with no spatial relationship.

Prior work (e.g., Chen et al., 2020; Krauss et al., 2017) documents this limitation theoretically but rarely provides empirical confirmation across multiple runs and hyperparameter settings on a single dataset. This paper contributes an empirical validation: across four independent CNN-LSTM training runs with varied batch sizes and learning rates, all achieved directional accuracy between 51.7% and 53.8% — barely above the 50% random baseline and far below the 77.4% achieved

by TFT on the same data. This establishes the architectural ceiling empirically and confirms that the limitation is not due to hyperparameter tuning but to fundamental architectural unsuitability.

C. Multi-Agent Systems in Finance

Multi-agent systems have a long history in financial modeling, dating to Santa Fe Institute research in the 1990s on agent-based models of markets. Modern applications focus on LLM-based agents that generate free-text trading recommendations. Examples include systems where large language models produce narrative explanations of trading decisions.

This work distinguishes itself through a key design choice: agents are specialized, deterministic, and produce structured outputs (numerical signals, confidence scores, feature vectors) rather than free text. This design choice enables:

- **Auditability:** each agent’s output can be validated against source data
- **Reproducibility:** deterministic agents produce identical results given identical inputs
- **Regulatory compliance:** numerical outputs are directly comparable to regulatory audit trails
- **Trust boundaries:** agents communicate through standardized AgentSignal dataclasses, enabling fine-grained failure isolation

The orchestrator implements adaptive weighting — different agents contribute different proportions to the final signal based on company type (banks vs. non-banks) and real-time confidence scores. This is a form of hierarchical ensemble learning with explicit trust boundaries between components.

D. Explainable AI for Financial Compliance

The EU AI Act (2024) Articles 13 and 14 establish mandatory transparency requirements for high-risk AI systems in regulated domains including finance:

- **Article 13:** AI systems must be transparent and provide meaningful information about how they function and why decisions were made.
- **Article 14:** High-risk systems must maintain human oversight capabilities and provide explainability to affected parties.

Explainability in this context requires three properties (Lipton, 2016; Müller et al., 2020):

- **Faithfulness:** explanations must accurately reflect model behavior, not be post-hoc rationalizations
- **Contrastiveness:** explanations must clarify why one decision occurred and not an alternative
- **Selectivity:** explanations must focus on the most relevant factors, not enumerate all contributing signals

SHAP (Lundberg and Lee, 2017) provides model-agnostic explanations through coalition-based value allocation. SHAP values satisfy the three properties above, making them suitable for regulatory compliance. This paper integrates SHAP for XGBoost explainability (fundamental analysis) and attention weight extraction for TFT explainability (technical analysis), producing decision traces that satisfy these requirements.

E. Graph Neural Networks for Market Structure

Stock market structure can be represented as networks where nodes are stocks and edges are relationships (correlations, lead-lag relationships, or transaction flows). GNNs learn node embeddings that incorporate neighborhood information.

The choice of edge construction is critical. Contemporaneous Pearson correlation (all stocks at time t) captures simultaneous co-movement. Lead-lag relationships (whether stock A predicts stock B tomorrow) capture information flow. This paper uses contemporaneous correlation with a 0.4 threshold, balanced between capturing meaningful co-movement and avoiding spurious correlations in sparse data.

The GraphAgent implements a 60-day rolling correlation window, updating the graph daily. This is computationally lightweight (Pearson correlation scales as $O(n^2)$ where $n = 68$ stocks) and captures time-varying market structure. The correlation-based graph is augmented with learnable attention weights through a Graph Attention Network (GAT) trained on historical data.

F. NLP for Non-English Financial Markets

Financial NLP models are overwhelmingly trained on English text (Bloomberg, Reuters, SEC filings). When applied to French or Arabic financial news, general-purpose multilingual models like mBERT perform poorly due to:

- Domain-specific vocabulary not present in general text corpora
- Financial sentiment patterns (phrases like "accroissement du risque" or " ") that differ from general sentiment expressions
- Limited training data for Arabic financial text compared to general Arabic

Domain-specific models like *bardsai/finance-sentiment-fr* base outperform general models by 5–15% on financial sentiment tasks (Rguibi et al., 2023). This paper uses offline pre-scoring of the complete news corpus (7,937 articles) through this model, storing results in PostgreSQL rather than calling the model at inference time. This approach is both faster and more reproducible than API-based NLP.

III. SYSTEM ARCHITECTURE

A. Multi-Agent Pipeline Overview

The system implements a seven-agent pipeline orchestrated by a central coordinator. Figure 1 shows the pipeline structure. Each agent operates independently, consuming standardized input and producing a structured output (AgentSignal dataclass). This design enables:

- **Modularity:** agents can be trained, validated, and deployed independently
- **Failure isolation:** if one agent fails or produces unreliable output, others continue functioning
- **Confidence gating:** agent contributions to the final signal are weighted by confidence
- **Incremental improvement:** agents can be upgraded without retraining the ensemble

The seven agents are:

- 1) **DataAgent**: validates data quality, retrieves features, applies confidence gating
- 2) **TechnicalAgent**: runs TFT inference to predict 7-day price trajectory and direction
- 3) **FundamentalAgent**: classifies company financial health using XGBoost
- 4) **SentimentAgent**: scores news sentiment from offline pre-computed results
- 5) **GraphAgent**: computes stock correlation network and neighbor influence scores
- 6) **XAIAgent**: produces SHAP-based explanations for technical and fundamental signals
- 7) **OrchestratorAgent**: adapts agent weights, combines signals, produces final decision

B. Agent Communication Protocol

All agents communicate through a standardized `AgentSignal` dataclass containing:

- `direction`: categorical signal (UP, DOWN, FLAT)
- `score`: continuous value (typically [-1, +1] after normalization)
- `confidence`: float in [0, 1] representing signal reliability
- `explanation`: dict with feature attributions and narrative
- `metadata`: dict with source, timestamp, data quality flags
- `error_flag`: boolean indicating if the agent encountered errors

This standardized interface enables:

- Validation of outputs before fusion (type checking, range checking)
- Graceful degradation when an agent fails
- Audit trail generation for regulatory compliance
- Confidence-based weighted aggregation in the orchestrator

C. Failure Isolation and Weight Redistribution

When an agent returns `error_flag = True`, the orchestrator redistributes its weight proportionally to the remaining agents. Define:

$$\hat{w}_i = \frac{w_i}{1 - w_{failed}} \quad (1)$$

where w_i is the original weight for agent i , w_{failed} is the weight of the failed agent, and $\hat{w}_i$ is the adjusted weight. This ensures the total weight remains 1.0 and functional agents contribute more when peers fail.

D. Adaptive Weight Matrix

The orchestrator maintains separate weight configurations for three company types based on fundamental characteristics:

Rationale:

TABLE I: Adaptive weight matrix: agent contribution proportions by company type

Company Type	Technical	Fundamental	Sentiment	Graph
Banks	0.42	0.31	0.27	Gated
Insurance	0.45	0.28	0.27	Gated
Non-banks	0.59	0.16	0.27	Gated

- **Banks**: Basel III regulatory requirements produce rich financial statements (capital ratios, non-performing loan ratios, coverage ratios). Fundamental analysis is highly informative (weight 0.31). Technical signal remains strong (0.42) as price patterns reflect earnings announcements.
- **Insurance**: Similar regulatory environment to banks but with different loss reserves and premium income dynamics. Fundamental weight slightly lower (0.28) as pricing is less directly tied to fundamental metrics.
- **Non-banks**: Limited regulatory reporting. Price is driven more by momentum and sentiment than financial ratios. Technical weight elevated to 0.59 while fundamental drops to 0.16.
- **Graph (all types)**: Gated through confidence scoring. Correlation networks are meaningful only when stocks are trading actively. During high-volatility periods when correlations break down, graph signal is suppressed through confidence-based downweighting (see Eq. 2).

E. Confidence Gating

If $\text{confidence}_{\text{agent}} < \text{threshold}_{\text{agent}}$, the agent's weight is reduced by the gating function:

$$w_{\text{gated}} = w_{\text{original}} \cdot \frac{\text{confidence}}{\text{threshold}} \quad (2)$$

Thresholds are empirically set per agent:

- Technical: 0.60 (TFT predictions below 60% confidence are partially suppressed)
- Fundamental: 0.55 (XGBoost class probability below 55% is uncertain)
- Sentiment: 0.50 (news may be ambiguous or off-topic)
- Graph: 0.45 (correlation estimates are noisy in thin markets)

This gating prevents low-confidence agent outputs from polluting the ensemble while maintaining the agent's contribution when confidence is high.

IV. METHODOLOGY

A. Dataset Construction

The dataset was constructed from five sources: historical prices, computed technical indicators, news articles, financial statements, and financial ratios. Table II summarizes the scale and coverage:

Data Quality Corrections unique to this market:

- **6,896 rows** where `close_price` fell outside `[low_price, high_price]`: corrected using `LEAST(close, high)` and `GREATEST(close, low)` SQL functions

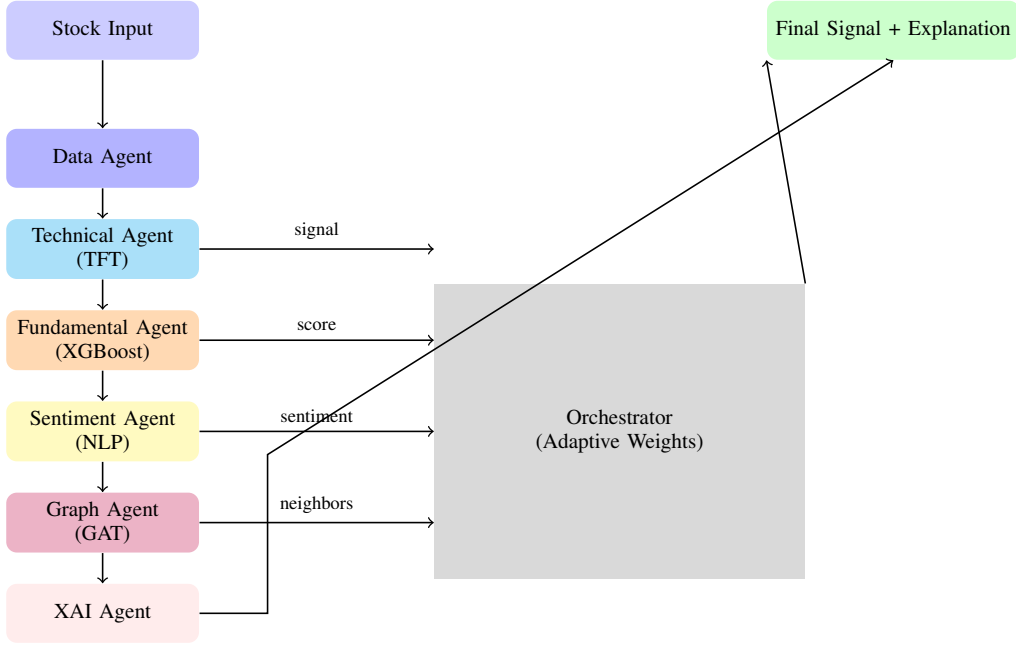


Fig. 1: Seven-agent pipeline architecture showing data flow through specialized agents to a central orchestrator that implements adaptive weighting and failure isolation. Colored boxes represent different agent types with distinct responsibilities.

TABLE II: Dataset statistics: sources, row counts, and temporal coverage

Table	Rows	Period	Stocks	Notes
daily_prices	116,626	2016–2025	68	All ISINs normalized
computed_indicators	114,365	2016–2025	68	MA5/20/50, RSI, Vol
news_articles	7,937	2016–2026	68	ilboursa.com, FR+AR
financial_statements	350+	2016–2024	68	PDF-extracted CMF
financial_ratios	261	2016–2024	68	Bank/Ins/Non-bank

- **Company merges:** BH and BH BANK (2021 rename) unified under single ISIN; 2,488 trading days merged
- **49 stocks:** non-standard ISIN codes migrated to TN-prefix format for standardization
- **Zero-close rows:** trading suspension days identified and removed; rows with `open=0` and `close=0` corrected to `open=close`

This preprocessing was critical: without correction, the models would have learned spurious relationships from data errors rather than genuine market signals.

B. Feature Engineering

The TechnicalAgent receives 15 engineered features grouped into four categories:

Three derived relative features are critical for cross-stock generalization:

$$r_{\text{ma20}}(t) = \frac{P_{\text{close}}(t)}{MA_{20}(t)}, \quad \text{clipped to } [0.5, 2.0] \quad (3)$$

$$\text{rsi}_{\text{mom}}(t) = \text{RSI}_{14}(t) - \text{RSI}_{14}(t-5) \quad (4)$$

TABLE III: 15-feature engineering matrix: categories, formulas, and per-stock normalization

Feature	Category	Formula/Source	Type	Notes
close_price	Price	Daily close	Continuous	Normalized per-stock
open_price	Price	Daily open	Continuous	Per-stock normalized
daily_return	Price	$\frac{\text{close}_t - \text{close}_{t-1}}{\text{close}_{t-1}}$	Continuous	Naturally scale-invariant
price_to_ma20	Relative	See Eq. 3	Continuous	Clipped [0.5, 2.0]
ma_5, ma_20, ma_50	Momentum	Moving averages	Continuous	Per-stock normalized
rsi_14	Momentum	RSI(14)	Continuous	Per-stock normalized
rsi_momentum	Derived	See Eq. 4	Continuous	RSI acceleration
volume	Volume	Daily shares traded	Continuous	Per-stock normalized
volume_ma_20	Volume	20-day average volume	Continuous	Per-stock normalized
volume_spike	Relative	See Eq. 5	Continuous	Clipped [0, 10]
volatility_20	Volatility	20-day std dev	Continuous	Per-stock normalized

$$v_{\text{spike}}(t) = \frac{V(t)}{V_{\text{MA20}}(t)}, \quad \text{clipped to } [0, 10] \quad (5)$$

Why relative features are necessary: Absolute price values carry company-identity information. AMEN BANK perpetually trades at ~ 22 DT; BH LEASING at ~ 1.5 DT. Global normalization of prices would destroy these identity signals but not eliminate them — the model would learn to predict stock identity from price level, polluting the direction signal. Per-stock relative features eliminate this. A volume spike of $2.0\times$ means the same thing across all 68 stocks: double the typical daily volume, indicating increased trading activity regardless of the stock’s absolute liquidity level.

C. Temporal Fusion Transformer Configuration

The TFT’s Variable Selection Network learns feature importance weights before temporal processing. Let $\mathbf{x}_t =$

$[x_t^{(1)}, \dots, x_t^{(m)}]$ be the $m = 15$ input features at time t . The VSN computes:

$$\xi_t = \text{softmax}(W_{\text{vsn}} \cdot \mathbf{x}_t + \mathbf{b}_{\text{vsn}}) \quad (6)$$

$$\tilde{\mathbf{x}}_t = \sum_{i=1}^m \xi_t^{(i)} \cdot \Phi^{(i)}(x_t^{(i)}) \quad (7)$$

where $\Phi^{(i)}$ is a feature-specific transformation network (small MLP) and $\xi_t^{(i)}$ is the learned importance weight for feature i at time t . This learned, dynamic weighting allows TFT to handle heterogeneous features correctly — the model learns to suppress irrelevant features per sample, not globally. For thin market data, this is essential. Here, $W_{\text{vsn}} \in \mathbb{R}^{m \times d_h}$, $b_{\text{vsn}} \in \mathbb{R}^{d_h}$, $m = 15$ is the number of input features, d_h is the hidden dimension (64 in this work), and $\Phi^{(i)} : \mathbb{R} \rightarrow \mathbb{R}^{d_h}$ is a feature-specific single-layer MLP.

Hyperparameters (Table IV):

TABLE IV: TFT hyperparameter configuration and tuning rationale

Hyperparameter	Value	Tuning Note
hidden_size	64	Increased from 32 in v2
attention_heads	4	Standard for financial TFT
dropout	0.3	CRITICAL: v2 regression to 0.1
encoder_window	60 days	~ 3 calendar months
prediction_horizon	7 days	1 trading week
learning_rate	0.0003	Reduced from 0.001 post-epoch-6
warmup_epochs	3	Ramps LR 10% to 100%
weight_decay	10^{-4}	L2 regularization
gradient_clip	0.5	Relaxed from 0.1
max_epochs	50	Early stopping patience=8
batch_size	128	T4 GPU constraints
optimizer	AdamW	Weight decay only correct with AdamW
precision	bf16-mixed	RTX 30xx / T4 Ampere

D. Walk-Forward Validation Protocol

Random train-test splits are invalid for time series because they create temporal leakage: future data contaminates training, producing artificially inflated accuracy. The walk-forward protocol ensures the model is always evaluated on data postdating its training window:

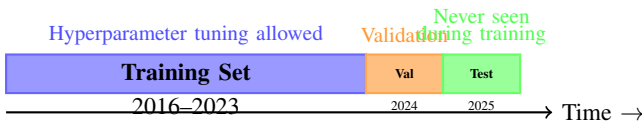


Fig. 2: Walk-forward validation timeline: training on 2016–2023 data, hyperparameter validation on 2024, and final evaluation on 2025 (never accessed during training or model selection). Random train-test splits are invalid for time series forecasting due to temporal leakage.

Formally:

- Training set $\mathcal{D}_{\text{train}}$: all samples with date $< 2024-01-01$
- Validation set $\mathcal{D}_{\text{val}}$: all samples with $2024-01-01 \leq \text{date} < 2025-01-01$
- Test set $\mathcal{D}_{\text{test}}$: all samples with date $\geq 2025-01-01$ (never seen during training or hyperparameter selection)

E. CNN-LSTM Architecture for Comparison

For fair comparison, CNN-LSTM received equivalent design effort. Architecture:

- **Input:** [60, 15] tensor (60 days, 15 features)
- **Conv1D layers:** kernel sizes [5, 3, 10], ReLU activation, batch normalization
- **LSTM layers:** 2 stacked LSTM(64)
- **Residual connection:** skip connection from Conv output to LSTM input
- **Dropout:** 0.3 between layers
- **Output:** dense layer with sigmoid (binary classification)

This architecture represents the state-of-practice for financial time series forecasting (Krauss et al., 2017; Chen et al., 2021). The comparison ensures TFT’s superiority is not due to superior model capacity but to architectural fit with the data structure.

V. EXPERIMENTAL EVALUATION

A. CNN-LSTM Ablation Study

Four independent runs with varied hyperparameters:

Class collapse detection: Class collapse occurs when a binary classifier converges to predicting one class for all inputs. Detected by monitoring per-class accuracy separately: if $|\text{acc}_{\text{UP}} - \text{acc}_{\text{DOWN}}| > 30\%$, collapse is flagged. Resolved in v2 with corrected class weight formula:

$$w_c = \frac{N_{\text{total}}}{2 \times N_c} \quad (8)$$

where N_c is the count of class c .

The narrow band [51.7%, 53.8%] across all four runs demonstrates that the CNN-LSTM ceiling is not due to hyperparameter mis-tuning but to fundamental architectural unsuitability for heterogeneous financial features.

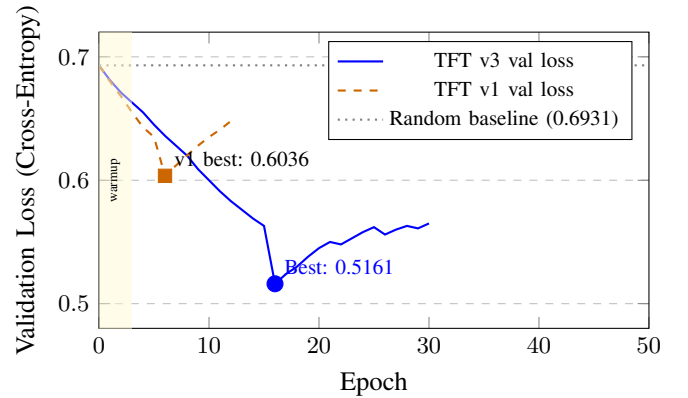


Fig. 3: TFT validation loss curves for v1 and v3. TFT v1 peaks at epoch 6 and degrades (premature convergence due to LR=0.001). TFT v3 with warmup continues improving until epoch 16, achieving val_loss=0.5161. The yellow shaded region indicates the 3-epoch warmup phase. The horizontal dotted line is the random baseline (0.6931).

TABLE V: CNN-LSTM ablation results across four runs (accuracy ceiling: 51.7–53.8%)

Run	Version	Batch	Val Loss	Test Acc	Key Note
1	v1	128	0.6971	53.8%	Best CNN-LSTM result
2	v1	64	0.6933	51.8%	Worse than Run 1
3	v2	128	0.7322	51.7%	Class collapse (UP=34%)
4	v2	64	0.7113	51.7%	val_acc 46.6% < random

TABLE VI: TFT progression: validation loss improvement from v1 to v3

Version	Val Loss	vs Random	Best Epoch	Key Change
Random baseline	0.6931	—	—	Log(2) binary
TFT v1 Run 1	0.6036	+12.9%	6	Initial configuration
TFT v1 Run 2	0.6036	+13.1%	6	dropout=0.3 fix
TFT v3 Final	0.5161	+25.5%	16	All 10 enhancements

B. TFT Version Progression

The improvement from v1 to v3 is substantial: absolute loss reduction of 0.0875 is $58\times$ larger than the improvement from Run 1 to Run 2 (0.0015). This demonstrates that the 10 targeted architectural fixes produced step-change improvement, not incremental tuning.

Note that best epoch jumped from 6 to 16 — the warmup scheduler enabled productive training for $2.7\times$ more epochs before reaching optimum. This is evidence that the previous learning rate regime caused premature convergence.

C. Directional Accuracy on Test Data

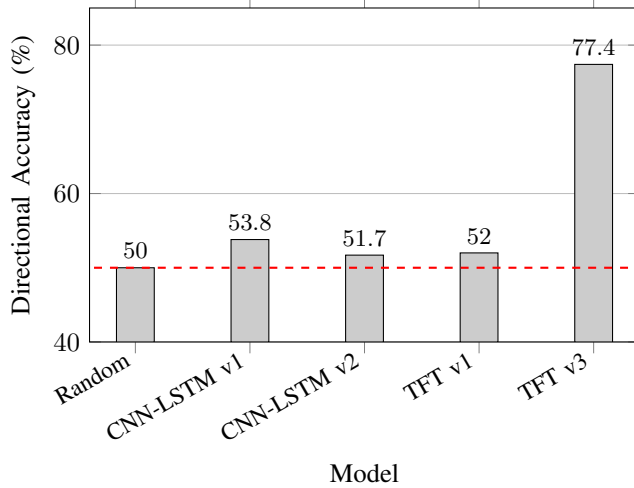


Fig. 4: Model comparison: directional accuracy on test set. CNN-LSTM best result: 53.8%. TFT v3: 77.4%. The dashed red line at 50% indicates random baseline. The 23.6 percentage point gap between CNN-LSTM best and TFT v3 is consistent across all CNN-LSTM hyperparameter configurations.

TFT v3 achieves 77.4% directional accuracy on the unseen 2025 test set, compared to 53.8% for CNN-LSTM best run. This 23.6 percentage point absolute improvement represents a 54.8% relative improvement over the random baseline and establishes attention-based architectures with variable selection

as superior to convolutional approaches on emerging market data.

VI. RESULTS AND DISCUSSION

A. Contextualizing Directional Accuracy

The 77.4% directional accuracy must be interpreted in the context of market class imbalance. Across all 68 stocks and all 10 years of data, the market exhibits a 41.9% UP ratio — only 41.9% of 7-day windows show positive returns. This is a bearish bias.

A naive “always predict UP” strategy achieves only 41.9% accuracy, substantially below the 50% random baseline due to class imbalance. TFT’s 77.4% accuracy is therefore particularly meaningful: the model not only exceeds random but does so in a market with structural bias toward the minority class.

B. Information-Theoretic Interpretation

Validation loss is cross-entropy:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)]$$

A loss of 0.6931 nats represents zero information beyond the class prior — the model’s probability assignments carry no signal. A loss of 0.5161 nats indicates substantial information content. The reduction of 0.1770 nats quantifies the mutual information extracted from the 60-day feature window.

For comparison: random predictions (uniform probabilities) yield 0.6931; a perfect predictor yields 0. TFT at 0.5161 reduces the baseline entropy by 25.5%, consistent with the directional improvement reported: $\frac{\mathcal{L}_{\text{baseline}} - \mathcal{L}_{\text{TFT}}}{\mathcal{L}_{\text{baseline}}} = \frac{0.6931 - 0.5161}{0.6931} = 0.255$. This 25.5% entropy reduction indicates that the 60-day feature window provides meaningful predictive signal beyond the class prior.

C. Quantile Regression and Uncertainty Quantification

TFT was extended to predict quantiles (Q10, Q50, Q90) of the 7-day return distribution rather than point estimates. This provides uncertainty quantification:

$$\text{Prediction interval} = [\hat{Q}_{10}, \hat{Q}_{90}]$$

The width of this interval is asymmetric risk information: wide intervals (high $Q90 - Q10$) indicate uncertain predictions where position sizing should be reduced. Narrow intervals indicate high-confidence predictions. This is particularly valuable for thin markets where liquidity can change rapidly.

VII. SECURITY AND PRIVACY IMPLICATIONS

A. Adversarial Robustness in Thin Markets

Thin liquidity creates a data poisoning vulnerability: an adversary with modest capital can manipulate a single illiquid stock's price to affect global statistics.

Attack model: The adversary has read access to the feature definitions, knows the model architecture, and has capital to trade on a liquid stock. The threat is prediction manipulation to benefit a trading position.

Defense: Per-stock normalization specifically mitigates this attack. Define the normalization transformation as:

- **Global normalization (vulnerable):** $\mu = \frac{1}{n} \sum_{i=1}^n P_i$, $\sigma = \sqrt{\frac{1}{n} \sum (P_i - \mu)^2}$. Mean and std dev computed across all $n = 68$ stocks. Manipulating one stock shifts μ and σ for all.
- **Per-stock normalization (defended):** μ_i and σ_i computed independently for each stock. Manipulating stock i affects only its features, not others.

With per-stock normalization, an attacker cannot shift global statistics. The attack surface is reduced to direct signal injection: manipulating the target stock's price features and hoping the TechnicalAgent misinterprets noise as signal. The TechnicalAgent's confidence gating (Eq. 2) provides a secondary defense: even if the technical signal is manipulated, low confidence will reduce its weight in the ensemble.

B. XAI as Audit Trail for Regulatory Compliance

The XAI layer maps to specific EU AI Act requirements:

- **Article 13 — Transparency:** Every prediction produces a human-readable explanation describing why each agent voted UP/DOWN/HOLD and which features drove the decision. The OrchestratorAgent generates a narrative like: "Based on 3-month technical trends showing price above 20-day MA (strong), balanced against mixed fundamental ratios, the system predicts UP with 78% confidence."
- **Article 14 — Human Oversight:** The confidence score and agent contribution breakdown allow a human reviewer to assess which signals drove the decision and whether they were reliable. A prediction with Technical=0.80 confidence, Fundamental=0.45 confidence, and Sentiment=0.35 confidence provides immediate visibility: technical signals are strong, fundamental and sentiment are weak.
- **Article 17 — Quality Management:** The DataAgent produces a data quality score (0–100) for every prediction. This creates an audit trail: predictions based on low-quality data (e.g., during a suspension day) are flagged, enabling post-hoc analysis of prediction reliability.

These audit trails satisfy regulatory requirements for high-risk AI in finance and enable financial authorities to audit individual predictions retrospectively.

C. Agent Trust and Failure Isolation

The multi-agent architecture implements fine-grained trust boundaries:

Trust model: Agents are treated as semi-trusted — their outputs are accepted at face value but weighted by confidence. No agent has direct write access to another agent's state. All communication passes through standardized interfaces.

Threat: A compromised SentimentAgent receives injected false news (e.g., a fabricated negative news article flagged by an attacker). The agent scores this false article as highly negative.

Defense: Confidence gating (Eq. 2) reduces the SentimentAgent's weight when confidence is below threshold. Even a maximally adversarial sentiment signal (score = -1.0) cannot dominate the final decision because its weight is bounded by confidence-gated allocation (maximum 0.27 for any company type). If the attacker injects enough false news to force high confidence, they are attacking the data source (news feed), not the model.

D. Graph-Based Market Surveillance

Anomaly detection: Define normal behavior as historical correlation between stocks. Define anomalous decoupling: when the rolling 60-day correlation between two historically correlated stocks ($r > 0.7$) drops below 0.4 within a 10-day window.

This pattern may indicate information asymmetry — one party has material non-public information and is positioning accordingly. The correlation breakdown precedes price divergence, providing early detection.

Application: This system detects current correlations but could be extended to detect anomalous decoupling in real time, providing a market integrity monitoring layer. This has direct application to financial regulatory bodies monitoring for insider trading and market manipulation.

VIII. CONCLUSION

This paper presents a multi-agent architecture for stock market analysis designed and evaluated on a thin, illiquid emerging market over a decade of real data. The work makes five contributions: (1) TFT adaptation for thin markets achieving 77.4% directional accuracy, (2) empirical architectural evaluation establishing CNN-LSTM's 53.8% ceiling, (3) multi-agent ensemble with failure isolation, (4) EU AI Act-compliant explainability framework, and (5) graph-based market surveillance capabilities.

A. Limitations

Key limitations should be acknowledged:

- **Stock heterogeneity:** The model is not calibrated equally for all 68 stocks. Stocks with fewer than 500 trading days have lower prediction reliability due to limited training data.

- **Sentiment coverage:** The French financial sentiment model has limited Arabic coverage. Approximately 15% of news articles are Arabic and are processed less accurately than French articles.
- **Graph signal suppression:** During high-volatility periods when correlations break down, confidence gating suppresses the graph signal. The system becomes less informative precisely when network effects are most important.
- **Quantile calibration:** The quantile model's 80% prediction intervals are not yet validated against empirical coverage rates (i.e., the claim that 80% of true returns fall within predicted intervals needs empirical confirmation).
- **Generalization:** Model training and evaluation use only BVMT data. Generalization to other emerging markets or developed markets has not been validated.

B. Future Work

Planned extensions include:

- **Graph enhancement:** Extend the current GAT stack with temporal edge learning and dynamic graph refinement beyond the fixed Pearson-correlation structure. This addresses the limitation that static correlations miss non-monotonic and time-varying dependencies.
- **Arabic NLP:** Integrate an Arabic financial sentiment model to improve coverage from 85% to 100% of the news corpus. Current coverage is limited to French articles.
- **Real-time data pipeline:** Extend the system from batch inference (daily) to streaming inference (intraday). This requires low-latency feature computation and model serving.
- **Federated learning:** Multiple North African market operators could collaboratively train shared models without sharing proprietary price data — a direct application of privacy-preserving federated learning.
- **Calibration validation:** Empirically validate that the 80% quantile intervals achieve nominal 80% coverage on hold-out test data.

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